



Bajaate Raho

A **self-hosted internet radio platform**, built from scratch to understand how audio streaming actually works.

● LIVE IN PRODUCTION

SWIPE TO SEE HOW IT'S BUILT →



THE MOTIVATION

Why build a streaming platform from scratch?

Using a music app is simple. **Building the thing that streams the audio** is not.

- How does audio stream over HTTP, not just download?
- How does a browser seek mid-playback, instantly?
- How do servers send part of a file, not the whole thing?

So I stopped reading about it — and built one.

The real production **topology** — not a diagram for show.



React Frontend

Material UI · HTML5 Audio API



GitHub Pages

Static frontend hosting



Cloudflare Tunnel

HTTPS, zero exposed ports



Nginx Reverse Proxy

Routes to localhost:8080



Spring Boot API

Java 21 · systemd service



Music Storage

Byte-range readable files

ENGINEERING DEEP DIVE

The hard part: HTTP Range Streaming

- 01 Browser requests a specific **byte range**, not the full file.
- 02 Server replies **206 Partial Content** with just that chunk.
- 03 Playback starts before the download finishes.
- 04 Seeking = a new range request — **never** a restart.

```
GET /stream/song.mp3
Range: bytes=1048576-2097151

HTTP/1.1 206 Partial Content
Content-Range: bytes 1048576-2097151/8388608
```

PRODUCTION DEPLOYMENT

Not localhost. A real Linux server, 24/7.

```
systemctl status -- bajaate-raho.service
```

```
ubuntu@bajaate-raho-server:~$ journalctl -u bajaate-raho -n 50
Aug 02 02:12:44 bajaate-raho-server systemd[1]: /etc/systemd/system/bajaate-rah>
Aug 02 02:12:44 bajaate-raho-server systemd[1]: Started bajaate-raho.service - >
lines 1-2/2 (END)
```

```
nginx.conf -- reverse proxy
```

```
location / {

    proxy_pass http://localhost:8080;

    proxy_http_version 1.1;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
```

Oracle Cloud VM

Ubuntu 24.04

systemd

Nginx

Cloudflare Tunnel



WHAT'S NEXT

Still a **work in progress** — that's the point.

- ☐ Migrate audio storage to object storage
- ☐ Detect and clean up duplicate songs
- ☐ Automated metadata processing pipeline
- ☐ Improve scalability under concurrent listeners